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SECTION A

1. In Technical drawing, a numerical linear dimension specified without a unit signifies
   A. foot (ft)
   B. Inches (in)
   C. Centimetres (cm)
   D. Millimetres (mm)

2. A wooden pencil sharpened conically should be rotated between the fingers while drawing a straight line, because:
   A. It is elegant
   B. It ensures a sharp and clear line
   C. It ensures a constant line thickness
   D. It ensures a neat drawing

3. Which of the following letter on a wooden pencil will signify the hardest lead?
   A. HB
   B. 3H
   C. 2H
   D. H

4. The ratio of thin lines to thick lines recommended for standard drawings is:
   A. 1:1
   B. 1:2
   C. 1:3
   D. 2:1

5. Which of the following drawing tool cannot qualify as a template?
   A. Sets of Circle
   B. 60/30 and 45 Set-squares
   C. Sets of Ellipse
   D. French curves

6. All title blocks must contain at least one of the following information ***EXCEPT***
   A. Title of drawing
   B. Names and dates of the drafter, checker and designer
   C. Dimensions
   D. Scale

7. The ratio of the size of the drawing of a part or machine to its actual size irrespective of the unit of measurement used is called a/an
   A. Sketch
   B. Scale
   C. Drawn size
   D. Actual size

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8. A circle is a plane figure bounded by the circumference that is always equidistant from a fix point known as.....

A. chord
B. radius
C. segment
D. centre
E. quadrant.

9. Which type of line has precedence over all other types of lines?

A. Continuous Thick Line
B. Centre Line
C. Continuous Thin Line
D. Hidden Line
E. All of the above

10. A/An ..... is created by the motion of a point on a circle as the circle rolled along a straight line

A. cycloid
B. epicycloid
C. hypocycloid
D. spiral
E. none of the above.

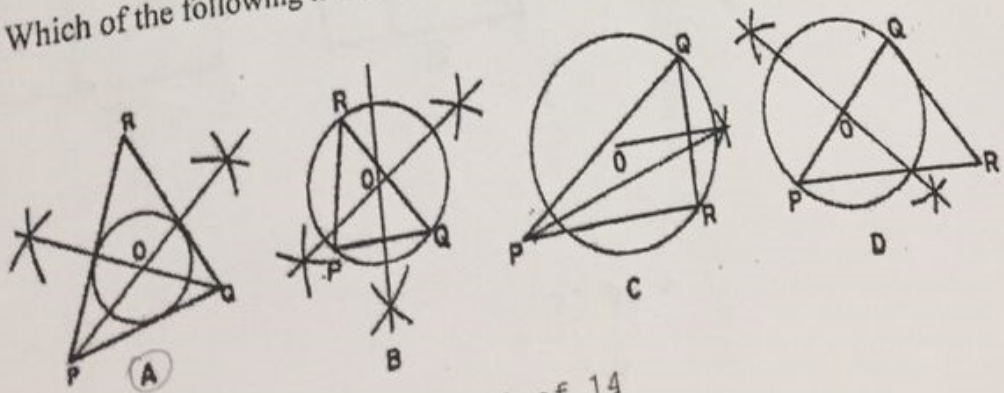
20mm = 2cm

11. If a length of 1metre on land is represented by 2cm on a drawing, what is the ratio of the scale used?

1m = 20m

A. 1:20
B. 1:200
C. 1:40
D. 1:400
E. 1:50

12. Which of the following is a circumscribed circle?



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13. The orthographic projection shown in Figure Q 13(b) of the object shown in Figure Q 13 (a) is in.....

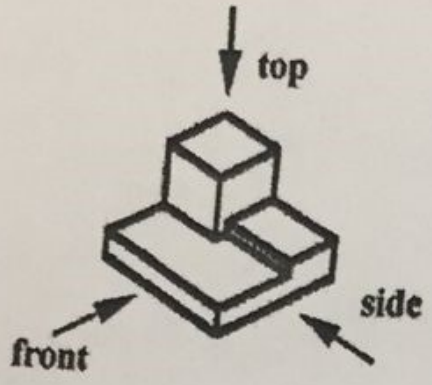


Figure Q 13(a)

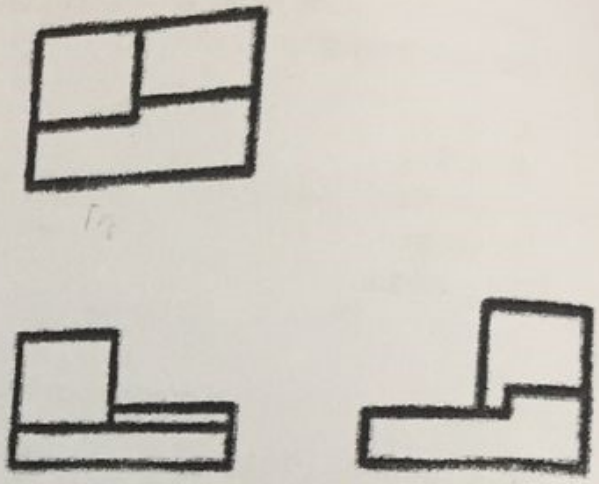
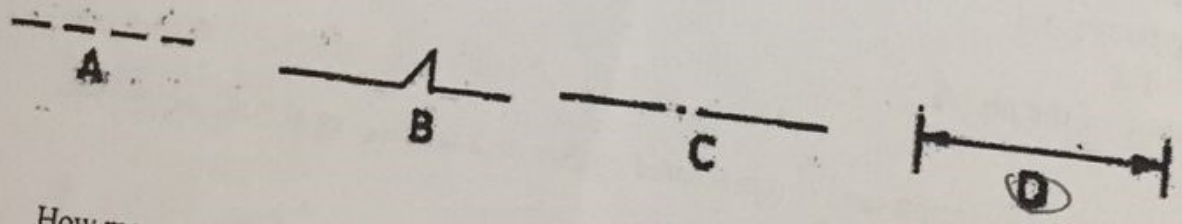


Figure Q 13(b)

A. First angle projection
B. Second Angle Projection
C. Third Angle Projection
D. None of the above

14. Which of the following shows a centre line?



15. How many ***pairs*** of parallel lines are there in a regular Hexagon?

A. 1
B. 2
C. 3
D. 6
E. None of the above

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Answer questions 16 to 17 using Figure 3

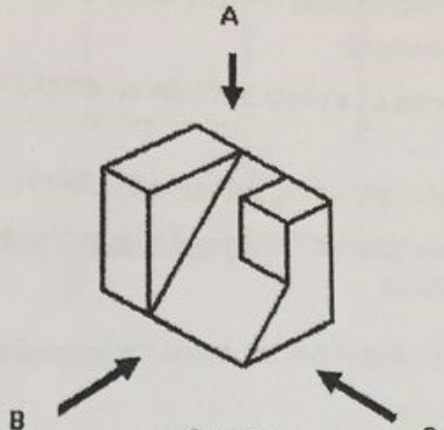
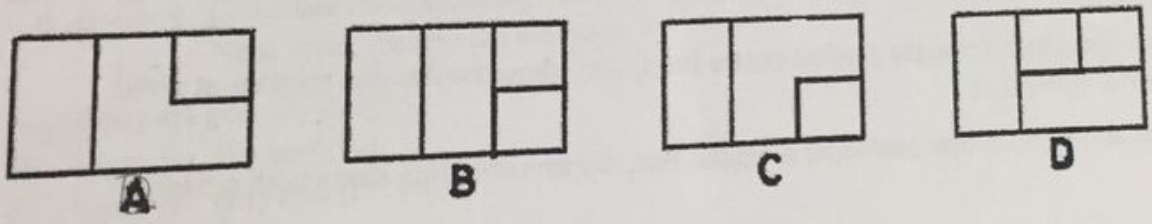
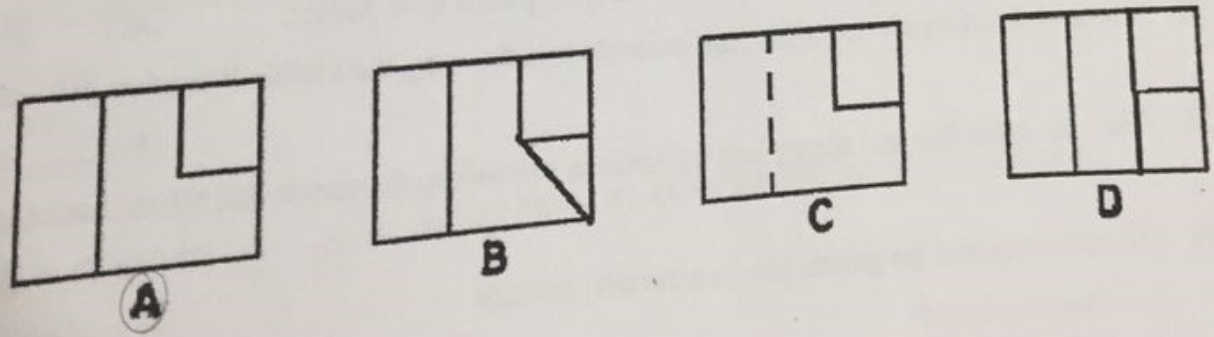


Figure 3

16. The front elevation of figure 3 in the direction of arrow B is .....

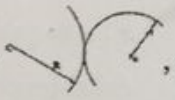


17. The plan of figure 3 in the direction of arrow A is .....



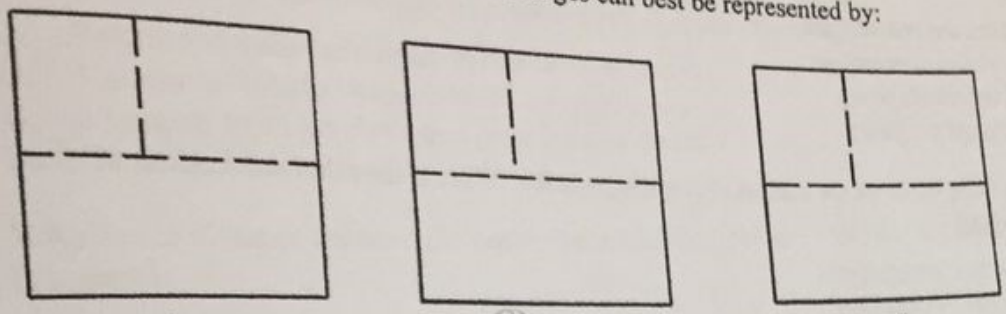
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Indicate whether the following statements from Q18 to Q34 are TRUE (T) or False (F).

18. A T-square in conjunction with a set square is required for horizontal lines. (A: T) B: F
19. The Path traced by a point attached within a circle that rolls on the outside of another circle is called inferior hypotrochoid. (A: T) B: F
20. The Path traced by a point on a circle that rolls on a straight line is called inferior cycloid. (A: T) B: F
21. Epitrochoids, Involutes of polygons, Spirals and Helices are plane curves (A: T) B: F
22. On the basis of Distance between Object and Ray, projections can be termed as perspective and parallel ray projections. (A: T) B: F
23. The symbol,  , signifies a 3rd Angle Orthographic Projection (A: T) B: F
24. The two arcs,  , are internally tangential (A: T) B: F
25. Hidden outlines are indicated in drawings with thick continuous lines. A:T (B: F)
26. Only line type A (thick continuous) and line type D (thin short dashes) represent edges. A:T (B: F)
27. The lengths and spacing of the elements of thick chain lines should differ from those of thin chain lines. A:T (B: F)
28. When thin chain lines are used as centre lines, they should cross one another at solid portions of the line. A:T (B: F)
29. When an invisible line continues a visible line, the invisible line starts with a dash. (A: T) B: F
30. All chain lines, E, should start and finish with a short dash. (A:T) B: F
31. Dashed lines should meet at corners and tangent points with dashes. A:T (B: F)
32. Where visible line and invisible line coincide in a drawing, the visible line takes precedence. (A: T) B: F
33. When invisible line and centre line coincide in a drawing, the centre line takes precedence. A: T B: F
34. The tools required for geometric construction include
    A. Drawing board
    B. T-square
    C. Pair of compasses
    D. All the above

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35. In a drawing, a meeting of two invisible edges can best be represented by:



A B C

36. All the following are plane figures except
    A. Polygons
    B. Cones
    C. Circles
    D. Triangles

37. For constructional purposes, which of the following conditions are necessary?
    i. Ensure thin constructional lines (well sharpened pencil),
    ii. Use large radii in the construction,
    iii. Consider how accuracy can be enhanced,
    iv. Neat work (minimize erasing)

    A. Only (i) and (ii) only
    B. Only (i), (ii) and (iv)
    C. Only (i), (iii) and (iv) only
    D. All

38. The intersection of the perpendicular bisectors of two chords gives a/an.....
    A. Circle
    B. Arc
    C. Centre
    D. Radius

39. A line is said to be .....to another line if it touches the line but does not cross it.
    A. Concentric
    B. Parallel
    C. Perpendicular
    D. Tangential

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40. The distance between the centres of two circles is R+r. What is the relationship between the two circles?

A. External intersection
B. Internal intersection
C. External tangency
D. Internal tangency

41. The distance between the centres of two circles is R-r. What is the relationship between the two circles?

A. External intersection
B. Internal intersection
C. External tangency
D. Internal tangency

42. For an external tangential relationship between two circles, which of the following statements is true:

A. The point of tangency and the centres of the two circles lie on a semi-circle
B. The point of tangency and the centres of the two circles form a right triangle
C. The point of tangency and the centres of the two circles must lie on a straight line
D. The point of tangency and the centres of the two circles lie on a circle

43. Which of the following curves is not a conic section?

A. Ellipse
B. Parabola
C. Hyperbola
D. Epicycloid

44. An accurate ellipse was required. Which of the methods should be applied?

A. Four centre method,
B. Rhombus method,
C. Concentric circle method
D. None of the above

45. The semi-circle (Right-triangle) theorem is applied in Principles of Tangency to construct a tangent to an arc or circle through a point

A. On the circle/ arc
B. Within the circle / arc
C. Outside the circle / arc
D. None

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46. The semi-circle (Right-triangle) theorem is a principle that can be applied to construct ...
    A. A common exterior tangent to two unequal circles
    B. A common interior tangent to two unequal circles
    C. A tangent to an arc through a point outside the arc
    D. All of the above

47. What is the distance between the centres of two circles (R=50, r =20) that are externally tangential.
    A. 30
    B. 35
    C. 50
    D. 70

48. What is the distance between the centres of two circles (R= 70, r = 20) that intersect one another.
    A. 30
    B. 40
    C. 50
    D. None of the above

49. The following are types of plane curves
    A. Cycloidal curves
    B. Involutes
    C. Spirals
    D. All of the above

50. A/an ..... is generated by a point on the circumference of a circle that rolls along a fixed straight line.
    A. Epicycloid
    B. Trochoid
    C. Cycloid
    D. All of the above

51. A/an ..... is generated by a point on the circumference of a rolling circle which rolls outside a directing circle.
    A. Epicycloid
    B. Trochoid
    C. Cycloid
    D. All of the above

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52. A /An..... is generated by a point within or outside the circle that rolls along a straight line.

A. Cycloid
B. Trochoid
C. Involute
D. Spiral

53. A/An ..... is a curve traced out by a point on a piece of string when unwound from any geometrical object.

A. Hypocycloid
B. Involute
C. Circle
D. Trochoid

54. The curve generated by a point moving continuously in one direction along a rotating line is called .....

A. Involute
B. Spiral
C. Cycloid
D. Trochoid

55. ..... is generated by a point within or outside a circle which rolls on the inside of the directing circle.

A. Hypotrochoid
B. Hypocycloid
C. Epitrochoid
D. Trochoid

56. Selection of views in Orthographic Projection are guided by the following rules. Which rule is considered first?

A. The rule of the longest side
B. The rule of stability
C. The rule of the most natural position
D. The rule of minimum dash lines or hidden details

57. In the first angle orthographic projection, the object is imagined to be placed

A. Below the HP and behind the VP
B. Above the HP and in front of the VP
C. Above the HP and behind the VP
D. None of the above

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58. A point has been projected onto the VP and HP as shown in Figure Q58. To which plane is the point closer?

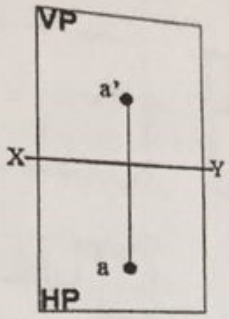


Figure Q58

A. VP-Vertical plane
B. HP- Horizontal plane
C. PP- Profile plane
D. AVP- Auxiliary vertical plane

59. Of all the types of projections, orthographic projection has been of prior importance in describing objects to the engineer. Which of the following characteristics makes orthographic projection exceptional to the engineer?

A. It is drawn to scale
B. It is drawn to the same shape
C. It can be dimensioned
D. It has self-correcting mechanisms

60. Figure Q60 shows a pictorial view of a block. Select a view from the views on the ***right*** of the figure that correctly represents the Front View of the object as indicated by the arrow.

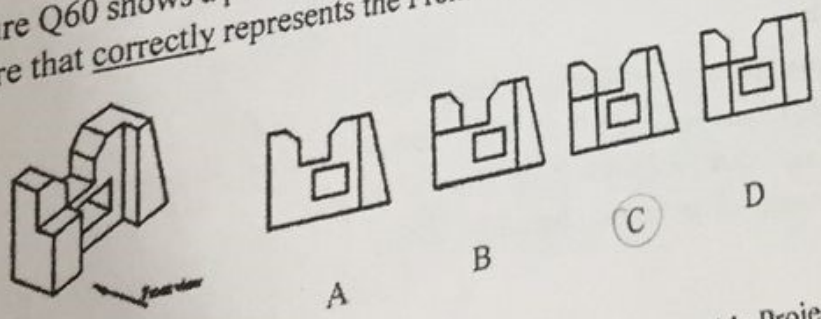


Figure Q60

61. The first most important view of an object during Orthographic Projection is captured on:

A. Vertical auxiliary plane
B. Horizontal plane
C. Vertical plane
D. Auxiliary plane

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Figure 4 shows a pictorial view of a block. The right side of the figure shows correctly projected views (i, ii, iii and iv) of the object. Use this Figure 4 to answer questions 62-65.

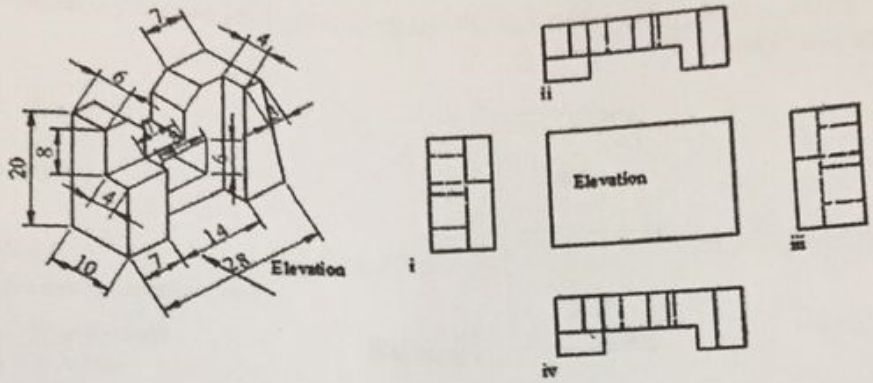


Figure 4

62. The view i. represents:

A. First angle orthographic projection Right Side View
B. First angle orthographic projection Left Side View
C. Third angle orthographic projection Right Side View
D. Third angle orthographic projection Left Side View

63. The view ii. represents:

A. First angle orthographic projection Top View
B. First angle orthographic projection Bottom View
C. Third angle orthographic projection Top View
D. Third angle orthographic projection Bottom View

64. The view iii. represents:

A. First angle orthographic projection Right Side View
B. First angle orthographic projection Left Side View
C. Third angle orthographic projection Right Side View
D. Third angle orthographic projection Left Side View

65. The view iv. represents:

A. First angle orthographic projection Top View
B. First angle orthographic projection Bottom View
C. Third angle orthographic projection Top View
D. Third angle orthographic projection Bottom View

OLOGY, KUMASI

IONS, 2021

TIME: 2 hours

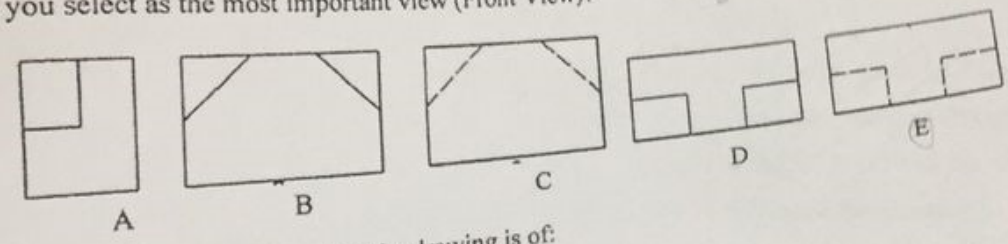
Title Block.

0.

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66. The following are properties of Orthographic projection *EXCEPT*:
   A. The rays are orthogonal to the planes of projection
   B. Only one side or surface is exposed to the rays
   C. The rays are parallel to one another
   D. It is a single-view projection

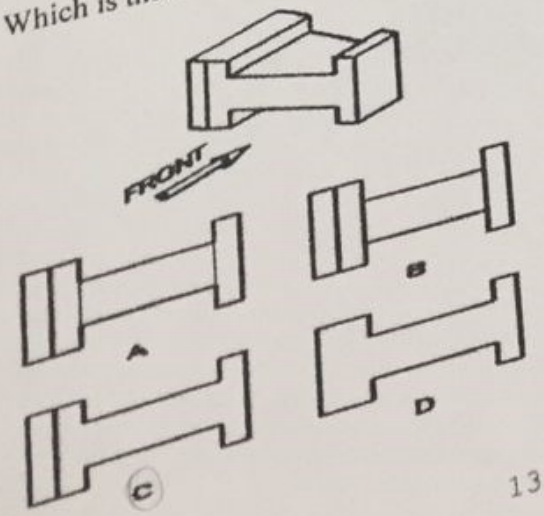
67. The orthographic views of a 3-dimensional object are shown below. Which of the views will you select as the most important view (Front View)?



68. In a reduction scale drawing, the drawing is of:
   A. The same size as the object
   B. Reduced size of the object
   C. Bigger size as the object
   D. Only some area that is reduced

69. A student is requested to purchase clutch (mechanical) pencils for producing engineering drawings. Which combination sizes of pencils will suit this purpose?
   A. 0.7 and 0.35
   B. 0.7 and 0.5
   C. 0.5 and 0.35
   D. 0.7 and 0.3

70. Which is the correct **FRONT** view of the figure below?



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SECTION B **(30 Marks)**

Provide the three most important First Angle Orthographic views of the object shown in **Figure QC** at scale of 1:1 on the drawing sheet.

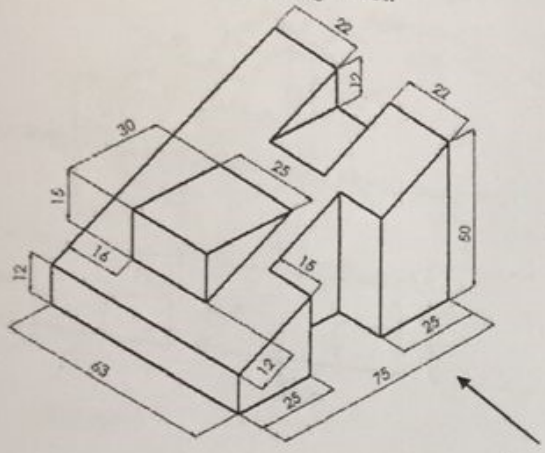
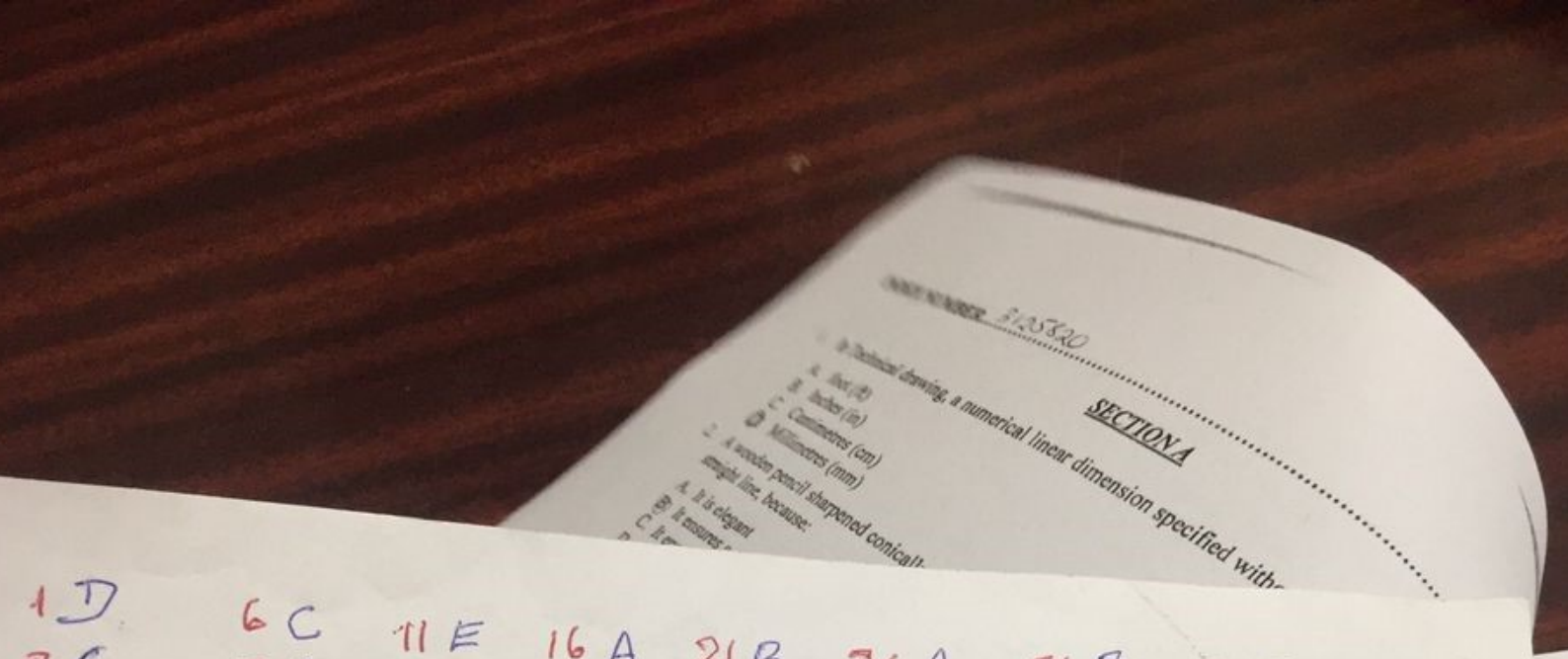


Figure QC

J. Aveyire
F. Davis
S. Boahen
F.K. Nyarko
D.E.K. Dzebre

A.K. Sunnu
G.F.K. Tay
C.K.K. Sekyere
P.O. Tawiah
E.A. Asante



1 D
2 C
3 B
4 B
5 B
6 C
7 B
8 D
9 A
10 A
11 E
12 B
13 C
14 C
15 C
16 A
17 A
18 B
19 B
20 B
21 B
22 A
23 B
24 B
25 B
26 A
27 B
28 A
29 A
30 B
31 B
32 A
33 A
34 C
35 A
36 B
37 D
38 C
39 D
40 C

41 D
42 C
43 D
44 C
45 C
46 D
47 D
48 D
49 D
50 C
51 A
52 B
53 B
54 B
55 A
56 A
57 B
58 B
59 D
60 C
61 C
62 D
63 C
64 C
65 A
66 D
67 B
68 B
69 A
70 C

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SECTION A

Questions Q1 to Q7: True or False

1. Technical drawing is expected to be: a) True b) False
   a. Unambiguous and clear,
   b. Complete,
   c. Suitable for duplication,
   d. Language-independent, and
   e. Conform to standard.
2. Technical drawing is a specification that is legal binding. a) True b) False
3. Technical drawing involves the following principles ONLY: a) True b) False
   a) standards b) Geometric
4. The BS standard format is International Standard. a) True b) False
5. BS EN ISO 5457 is an adopted standard. a) True b) False
6. Inherent errors can be completely avoided by computerised draughting. a) True b) False
7. The diameter of the circumscribed circle of an odd regular polygon is equal to the distance between flats (BF), a) True b) False

Questions Q8 to Q30: **Fill-In** or **Multiple-Choice Questions**

8. List the three creation techniques employed in Technical Drawing
   a) Free hand Sketch b) Mechanical/ Using instrument c) Computerized// Graphical
9. Which of the following (one) do not form part of the basic information in a title block?
   a. Name of company or organisation,
   b. Title of drawing,
   c. Drawing number,
   d. Date with name of staff who designed, drawn, checked, and
   e. None
10. The scale ratio specified in the Title Block is:
    a) Engineers Scale b) Graphical Scale
11. A scale ratio of 2:1 signifies:
    a) Reduction scale b) Magnification scale c) Full size
12. The maximum length is 50 m and length of scale is 10 cm. Find the Representative factor.
    a) 1/50 b) 50 c) 500 d) 1/500
13. Given Representative factor as 4/50 and maximum length is 100 cm. Find the length of scale.
    a) 8 cm b) 10 cm c) 9.7 cm d) 12 cm
14. Which of the following is a conic section?

a) Circle b) Rectangle c) Triangle d) Square

15. The sections cut by a plane on a right circular cone are called as _____
a) Parabolic sections b) Conic sections c) Elliptical sections d) Hyperbolic sections

16. While cutting, if the plane is at an angle and it cuts all the generators, then the conic formed is called as _____
a) Circle b) Ellipse c) Parabola d) Hyperbola

17. The locus of point moving in a plane such that the distance between a fixed point and a fixed straight line is constant is called as _____
a) Conic b) Rectangle c) Square d) Polygon

18. Which of the following scales is used to accurately measure minute divisions?
a) Plain scale b) Diagonal scale c) Scale of chords d) Comparative scale

19. How many tangents can be drawn from a point outside a given circle?
a) 4 b) 3 c) 2 d) 1

20. In how many ways can there be a common tangent between two circles?
a) 3 b) 4 c) 1 d) 2

21. Which of the following is incorrect about Ellipse?
a) Eccentricity is less than 1
b) Mathematical equation is $X^2/a^2 + Y^2/b^2 = 1$
c) If a plane is parallel to axis of cone cuts the cone, then the section gives ellipse
d) The sum of the distances from two focuses and any point on the ellipse is constant

22. Which of the following is the eccentricity for an ellipse?
a) 1 b) 3/2 c) 2/3 d) 5/2

23. Steps are given to draw an ellipse by loop of the thread method. Arrange the steps.

i. Check whether the length of the thread is enough to touch the end of minor axis.
ii. Draw two axes AB and CD intersecting at O. Locate the foci F1 and F2.
iii. Move the pencil around the foci, maintaining an even tension in the thread throughout and obtain the ellipse.
iv. Insert a pin at each focus-point and tie a piece of thread in the form of a loop around the pins.

a) i, ii, iii, iv b) ii, iv, i, iii c) iii, iv, i, ii d) iv, i, ii, iii

24. Which of the following does not belong to ellipse?
a) Latus rectum b) Directrix c) Major axis d) Asymptotes

25. In the general method of drawing an ellipse, a vertical line called as ____ is drawn first.
a) Tangent b) Normal c) Major axis d) Directrix

26. If eccentricity of ellipse is 3/7, how many divisions will the line joining the directrix and the focus have in general method?
a) 10 b) 7 c) 3 d) 5

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27. The semi-circle (Right-triangle) theorem is applied in Principles of Tangency to construct a tangent to an arc or circle though a point .....:
a) On circle/ arc, b) Within circle/ arc, c) Outside circle / arc, d) None

28. The semi-circle (Right-triangle) theorem is a principle that can be applied to construct a _____
a) Common exterior tangent to two unequal circles,
b) Common interior tangent to two unequal circles,
c) Tangent to an arc through a point outside the arc,
d) All of the above.

29. What is the distance between the centres of two circles (R=40, r=10) that are external tangential?
a) 30 b) 40 c) 50 d) 60

30. What is the distance between the centres of two circles (R=40, r=10) that are internal tangential?
a) 30 b) 40 c) 50 d) None